Worked solutions to

Bayesian Statistical Methods

With Applications to Machine Learning — Reich & Ghosh (2nd ed.)

Chapter 9

Statistical properties of Bayesian methods

Every exercise is presented in the same order:

- the **exact** exercise text from the book;
- the problem **in plain words**;
- the **background** you need, recalled from scratch;
- the **key idea** that unlocks the problem;
- a **step-by-step solution** — every algebra step written out, with a *Why?* note whenever a step is not obvious;
- a **sanity check** of the answer;
- the boxed **answer**;
- **takeaways** and common mistakes.

All numbers and figures are reproduced by `ch09/code/ch09.R` (base R, under one second).

Sai Sridhar · October 21, 2026

Contents

Answers at a glance 2
Primer: the tools of this chapter 3
Exercise 9.1 — Choosing the rejection threshold 4
Exercise 9.2 — Should the company run the confirmatory trial? 6
Exercise 9.3 — Bias–variance tradeoff for a proportion 8
Exercise 9.4 — Posterior mean vs. posterior mode for a variance 10
Exercise 9.5 — Frequentist coverage of Jeffreys credible intervals 12
Exercise 9.6 — A Bayesian re-analysis from a published estimate 14
Exercise 9.7 — How many customers for a posterior SD of 0.01? 16
Exercise 9.8 — Exponential families, conjugacy and unbiasedness 17

Answers at a glance

1	$\mu \mid Y \sim \mathrm{N}(Y/2, 1)$; $c = 0.5$ (equal costs), $c = 10/11$ (type I ten times worse); type I error $= 1 - \Phi(\sqrt{2}\,\Phi^{-1}(c))$; $c = 0.878$ gives 0.05.
2	Run the trial iff $c\{0.8\theta + 0.05(1-\theta)\} > 1$, i.e. $\theta > \frac{1/c - 0.05}{0.75}$ (never if $c \leq 1.25$).
3	$\hat{\theta}_1$ unbiased, $\hat{\theta}_2$ biased toward $\frac{1}{2}$ but less variable; $\hat{\theta}_2$ wins iff $\theta_0(1-\theta_0) > \frac{n}{12n+4}$ (roughly $0.09 < \theta_0 < 0.91$).
4	Mean $\frac{b+\sum Y_i^2/2}{a+n/2-1} = 24.99$, mode $\frac{b+\sum Y_i^2/2}{a+n/2+1} = 16.66$; both $\to \sigma^2$.
5	Jeffreys intervals have roughly nominal *average* coverage (0.95–0.96 for $n \geq 5$) but fluctuate (0.89–0.99) because Y is discrete.
6	SE $= 0.70$; flat prior: $\beta_1 \mid \text{data} \approx \mathrm{N}(1.8, 0.70^2)$, $\mathrm{Prob}(\beta_1 > 0) = 0.995$; sceptical priors shrink it.
7	$n \approx \theta(1-\theta)/0.01^2 - 3 = 2{,}397$.
8	$\psi = \eta^2/2$ (normal), $\psi = e^\eta$ (Poisson); MLE $\bar{Y}$ unbiased; conjugate $\pi(\eta) \propto e^{n_0 y_0 \eta - n_0 \psi(\eta)}$; Bayes estimator $\frac{n_0 y_0 + n\bar{Y}}{n_0 + n}$, never unbiased for a proper prior.

A five-minute primer on the tools of this chapter

Background you need

1. Decision theory. Choose the action a minimising the *posterior expected loss* $\mathrm{E}[L(\theta, a) \mid \mathbf{Y}]$. For a test with loss c_I for a type I error (rejecting a true H_0) and c_{II} for a type II error, reject H_0 iff

$$c_I \,\mathrm{Prob}(H_0 \mid \mathbf{Y}) < c_{II} \,\mathrm{Prob}(H_1 \mid \mathbf{Y}) \iff \mathrm{Prob}(H_1 \mid \mathbf{Y}) > \frac{c_I}{c_I + c_{II}}.$$

Under squared-error loss the Bayes estimator is the posterior mean.

2. Frequentist properties. Fix the true parameter θ_0 and imagine repeating the experiment. For an estimator $\hat{\theta}$: bias $= \mathrm{E}(\hat{\theta}) - \theta_0$, and MSE $=$ bias$^2 + \mathrm{Var}(\hat{\theta})$. For an interval: coverage $= \mathrm{Prob}(\theta_0 \in \text{interval})$.

3. Asymptotics. With lots of data the posterior is approximately $\mathrm{N}(\hat{\theta}_{\mathrm{MLE}},\ \mathrm{SE}^2)$ whatever the (smooth, positive) prior — the Bernstein–von Mises theorem. This lets us do a Bayesian analysis from a reported estimate and standard error.

4. Exponential families. $f(y \mid \eta) \propto \exp\{y\eta - \psi(\eta)\}$ has $\mathrm{E}(Y) = \psi'(\eta)$ and $\mathrm{Var}(Y) = \psi''(\eta)$; the conjugate prior has the same form in η.

Exercise 9.1 — Choosing the rejection threshold

Exercise 1 (exact text from the book)

Assume $Y|\mu \sim \text{Normal}(\mu, 2)$ and $\mu \sim \text{Normal}(0, 2)$ (i.e., $n = 1$). The objective is to test the null hypothesis $H_0 : \mu \leq 0$ versus the alternative hypothesis that $H_1 : \mu > 0$. We will reject H_0 if $\text{Prob}(H_1|Y) > c$.

(a) Compute the posterior of μ.

(b) What is the optimal value of c if Type I and Type II errors have the same costs?

(c) What is the optimal value of c if a Type I error costs ten times more than a Type II error?

(d) Compute the Type I error rate of the test as a function of c.

(e) How would you pick c to control Type I error at 0.05?

In plain words

One normal observation, a normal prior, and the rule "reject if the posterior probability of $\mu > 0$ exceeds c". Choose c from costs, then study how often the rule rejects a true null.

Background you need

Normal–normal update (Chapter 2 primer); decision rule for tests (primer, point 1). Both variances here are 2 (the second argument of Normal is the variance).

Key idea

$\text{Prob}(H_1 \mid Y) = \Phi(Y/2)$ is increasing in Y, so "$\text{Prob}(H_1 \mid Y) > c$" is the same as "$Y >$ a cut-off"; the type I error is a normal tail probability at the boundary $\mu = 0$.

Solution

(a) Posterior.

Precisions $\frac{1}{2} + \frac{1}{2} = 1$; mean $\frac{Y/2+0}{1} = \frac{Y}{2}$. So $\mu \mid Y \sim \text{Normal}(Y/2,\ 1)$ and $\text{Prob}(H_1 \mid Y) = \text{Prob}(\mu > 0 \mid Y) = \Phi(Y/2)$.

(b) Equal costs.

$c = \frac{c_I}{c_I + c_{II}} = \frac{1}{2}$: reject when H_1 is more probable than H_0 (i.e. when $Y > 0$).

(c) Type I error ten times worse.

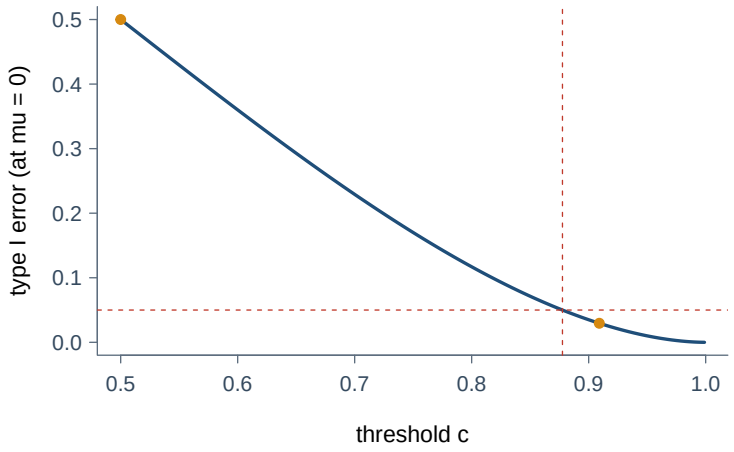
$c = \frac{10}{10+1} = \frac{10}{11} = 0.909$.

(d) Type I error.

Reject iff $\Phi(Y/2) > c \iff Y > 2\Phi^{-1}(c)$. Under H_0 the probability of rejection is largest at the boundary $\mu = 0$, where $Y \sim \text{N}(0, 2)$:

$$\alpha(c) = \text{Prob}\big(Y > 2\Phi^{-1}(c) \mid \mu = 0\big) = 1 - \Phi\Big(\frac{2\Phi^{-1}(c)}{\sqrt{2}}\Big) = 1 - \Phi\big(\sqrt{2}\,\Phi^{-1}(c)\big).$$

For example $\alpha(0.5) = 0.5$ and $\alpha(10/11) = 0.030$.



(e) Controlling type I error at 0.05.

Solve $\alpha(c) = 0.05$: $\sqrt{2}\,\Phi^{-1}(c) = 1.645 \Rightarrow \Phi^{-1}(c) = 1.163 \Rightarrow c = \Phi(1.163) = \mathbf{0.878}$.

Sanity check

10^6 simulated datasets with $\mu = 0$ and $c = 0.8776$ reject 5.01% of the time. ✓

Answer

(a) $\mu \mid Y \sim \mathrm{N}(Y/2, 1)$. (b) $c = 0.5$. (c) $c = 10/11 \approx 0.909$. (d) $\alpha(c) = 1 - \Phi(\sqrt{2}\,\Phi^{-1}(c))$ (worst case $\mu = 0$). (e) $c = \Phi(1.645/\sqrt{2}) = 0.878$.

Takeaway & common mistakes

A Bayesian threshold can be chosen from costs (decision theory) or calibrated to a frequentist error rate; here a 10:1 cost ratio already gives a type I error of only 3%.

Exercise 9.2 — Should the company run the confirmatory trial?

Exercise 2 (exact text from the book)

Given data from a small pilot study, your current posterior probability that the new drug your company has developed is more effective than the current treatment is $\theta \in [0, 1]$. Your company is considering to run a large clinical trial to confirm that your drug is indeed preferred. If you run the trial it will cost X. *If in fact your drug is better, then the probability that you will confirm this in the trial is 80%; if in fact …* cX. For which values of θ and c would you initiate the trial?

Note

In the book (and our copy) the middle of this exercise is garbled by a typesetting error: the text after "80%; if in fact" is lost. We use the natural reading: *if the drug is not better, the trial (wrongly) confirms it with probability 5%; if the trial confirms the drug, the company earns* cX. The general solution with a false-confirmation probability α is also given, so any other version of the numbers can be plugged in.

In plain words

Running the trial costs X; a successful (confirming) trial is worth cX. Given our current belief θ that the drug is better, is the gamble worth it?

Background you need

Expected value of a decision $=$ $\sum$(payoff) $\times$ Prob(outcome). Law of total probability: $\text{Prob}(\text{confirm}) = \text{Prob}(\text{confirm} \mid \text{better})\,\theta + \text{Prob}(\text{confirm} \mid \text{not better})\,(1-\theta)$.

Key idea

Run the trial iff its expected net gain is positive: $cX \cdot \text{Prob}(\text{confirm}) - X > 0$.

Solution

› Step 1 — probability of confirmation.

$\text{Prob}(\text{confirm}) = 0.8\theta + \alpha(1-\theta)$, with $\alpha = 0.05$.

› Step 2 — expected net gain.

$\text{E}(\text{gain}) = X\left[c\{0.8\theta + \alpha(1-\theta)\} - 1\right]$ (not running gains 0).

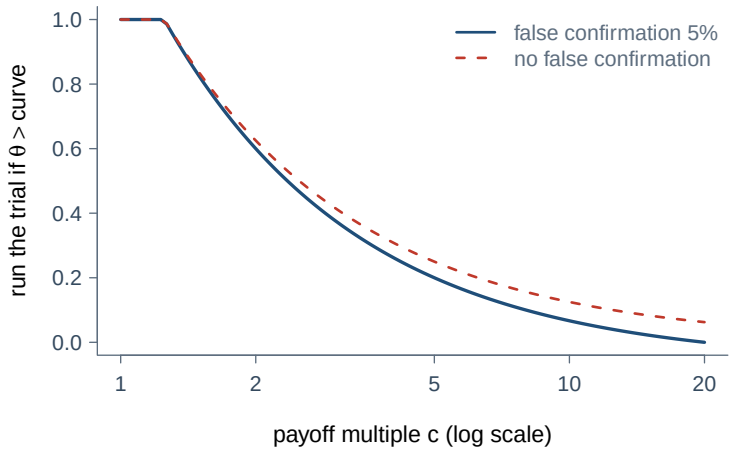
› Step 3 — decision rule.

Run iff $c\{0.8\theta + \alpha(1-\theta)\} > 1$, i.e.

$$\theta > \theta^*(c) = \frac{1/c - \alpha}{0.8 - \alpha} \qquad (= \tfrac{1/c - 0.05}{0.75} \text{ for } \alpha = 0.05).$$

Why? Divide by c and solve the linear inequality $0.75\theta + 0.05 > 1/c$ for θ.

c	1.25	1.5	2	5	20
run if $\theta >$	never (≥ 1)	0.822	0.600	0.200	always (≤ 0)



Without false confirmations ($\alpha = 0$) the rule is simply $0.8\,c\,\theta > 1$, i.e. $\theta > \frac{1.25}{c}$ (dashed curve).

Answer

Run the trial iff $c\{0.8\theta + 0.05(1-\theta)\} > 1$, i.e. $\theta > \frac{1/c - 0.05}{0.75}$. It is never worth it if $c \leq 1.25$ (even a certainly-better drug is confirmed only 80% of the time), and always worth it if $c \geq 20$; e.g. with $c = 2$ run it when $\theta > 0.6$.

Takeaway & common mistakes

Decision theory turns "how sure are we?" (θ) and "how much is at stake?" (c) into a single, transparent rule.

Exercise 9.3 — Bias–variance tradeoff for a proportion

Exercise 3 (exact text from the book)

Assume $Y|\theta \sim \text{Binomial}(n, \theta)$. Consider two estimators of θ: the sample proportion $\hat{\theta}_1(Y) = Y/n$ and the posterior mean under the Jeffreys prior $\hat{\theta}_2(Y) = (Y + 1/2)/(n + 1)$.

(a) Compute the bias, variance and MSE for each estimator, and comment on the bias-variance tradeoff.

(b) In terms of n and the true value θ_0, when is $\hat{\theta}_2(Y)$ preferred?

In plain words

Compare the plain proportion with the slightly shrunken Jeffreys estimate in repeated samples.

Background you need

$\text{E}(Y) = n\theta_0$, $\text{Var}(Y) = n\theta_0(1-\theta_0)$. For $\hat{\theta} = aY + b$: $\text{E}\hat{\theta} = an\theta_0 + b$, $\text{Var}\hat{\theta} = a^2 n\theta_0(1-\theta_0)$.

Key idea

$\hat{\theta}_2 = \frac{n}{n+1} \cdot \frac{Y}{n} + \frac{1}{n+1} \cdot \frac{1}{2}$: a weighted average of Y/n and $\frac{1}{2}$ — shrinkage toward $\frac{1}{2}$ trades a little bias for less variance.

Solution

(a) $\hat{\theta}_1 = Y/n$.

Bias 0; variance $\frac{\theta_0(1-\theta_0)}{n}$; MSE $= \frac{\theta_0(1-\theta_0)}{n}$.

(a) $\hat{\theta}_2 = \frac{Y+1/2}{n+1}$.

$$\text{bias} = \frac{n\theta_0 + \frac{1}{2}}{n+1} - \theta_0 = \frac{\frac{1}{2} - \theta_0}{n+1}, \qquad \text{Var} = \frac{n\theta_0(1-\theta_0)}{(n+1)^2}, \qquad \text{MSE} = \frac{n\theta_0(1-\theta_0) + (\frac{1}{2} - \theta_0)^2}{(n+1)^2}.$$

Comment: $\hat{\theta}_2$ is biased toward $\frac{1}{2}$, but its variance is smaller by the factor $\left(\frac{n}{n+1}\right)^2$. Both bias and the variance gap vanish as $n \to \infty$.

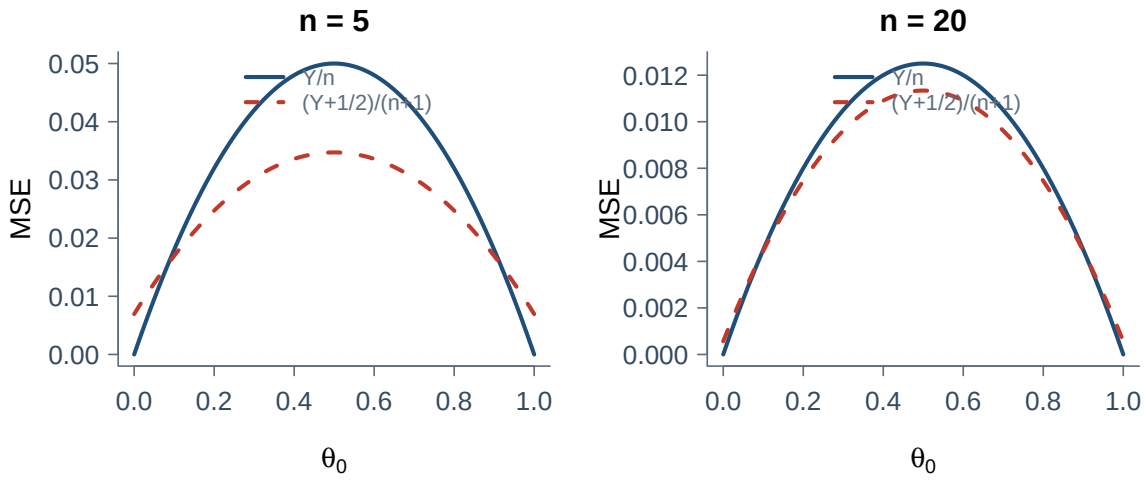
(b) When does $\hat{\theta}_2$ win?

Write $v = \theta_0(1-\theta_0)$ and note $(\frac{1}{2} - \theta_0)^2 = \frac{1}{4} - v$. Then $\text{MSE}_2 < \text{MSE}_1$ iff

$$n\big(nv + \tfrac{1}{4} - v\big) < (n+1)^2 v \iff \tfrac{n}{4} < (3n+1)v \iff \theta_0(1-\theta_0) > \frac{n}{12n+4}.$$

Why? Multiply both MSEs by $n(n+1)^2$, expand $(n+1)^2 v = n^2 v + 2nv + v$, and collect the v terms.

The bound increases to $\frac{1}{12}$ as $n \to \infty$, so $\hat{\theta}_2$ is better for θ_0 in about $(0.09,\ 0.91)$: $n = 5$: $(0.085, 0.915)$; $n = 20$: $(0.090, 0.910)$; $n = 100$: $(0.091, 0.909)$.



Answer

(a) $\hat{\theta}_1$: bias 0, MSE $\theta_0(1-\theta_0)/n$. $\hat{\theta}_2$: bias $\frac{1/2-\theta_0}{n+1}$, variance $\frac{n\theta_0(1-\theta_0)}{(n+1)^2}$, MSE $\frac{n\theta_0(1-\theta_0)+(1/2-\theta_0)^2}{(n+1)^2}$. (b) $\hat{\theta}_2$ has smaller MSE iff $\theta_0(1-\theta_0) > \frac{n}{12n+4}$, i.e. unless θ_0 is within about 0.09 of 0 or 1.

Takeaway & common mistakes

Unbiasedness is not optimality: a small, well-placed bias usually lowers the MSE.

Exercise 9.4 — Posterior mean vs. posterior mode for a variance

Exercise 4 (exact text from the book)

Assume $Y_1, \ldots, Y_n | \sigma^2 \overset{\text{iid}}{\sim} \text{Normal}(0, \sigma^2)$ and $\sigma^2 \sim \text{InvGamma}(a, b)$.

(a) Give an expression for two estimators of σ^2: the posterior mean and the posterior mode.

(b) Plot the posterior density functions, and then give the two estimators for a sample with $n = 10$ and $\sum_{i=1}^{n} Y_i^2 = 200$ and $a = b = 0.001$.

(c) Argue that both estimators are consistent (i.e., their MSE goes to zero as n increases) for any a and b.

In plain words

Two Bayesian point estimates of a normal variance: derive them, compute them for a sample, and show they both converge to the truth.

Background you need

$\sigma^2 \mid \mathbf{Y} \sim \text{InvGamma}(A, B)$ with $A = a + \frac{n}{2}$, $B = b + \frac{1}{2}\sum Y_i^2$ (Exercise 2.8). $\text{InvGamma}(A, B)$: mean $\frac{B}{A-1}$ ($A > 1$), mode $\frac{B}{A+1}$. $\sum Y_i^2 \sim \sigma^2 \chi^2_n$ (mean $n\sigma^2$, variance $2n\sigma^4$).

Key idea

Both estimators are "$\sum Y_i^2/2$ divided by roughly $n/2$", so both behave like the MLE $\frac{1}{n}\sum Y_i^2$ for large n.

Solution

(a) Formulas.

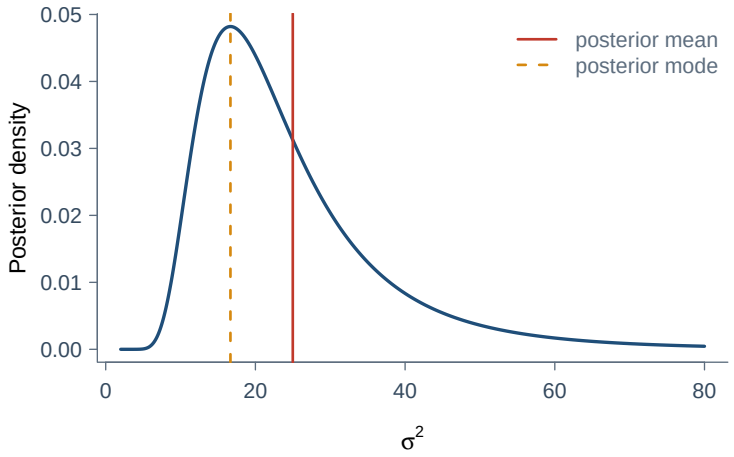
$$\hat{\sigma}^2_{\text{mean}} = \frac{b + \frac{1}{2}\sum Y_i^2}{a + \frac{n}{2} - 1}, \qquad \hat{\sigma}^2_{\text{mode}} = \frac{b + \frac{1}{2}\sum Y_i^2}{a + \frac{n}{2} + 1}.$$

Why? The mode of $x^{-A-1}e^{-B/x}$: set the derivative of $-(A+1)\log x - B/x$ to zero: $-\frac{A+1}{x} + \frac{B}{x^2} = 0 \Rightarrow x = \frac{B}{A+1}$.

(b) Numbers.

$A = 0.001 + 5 = 5.001$, $B = 0.001 + 100 = 100.001$:

$$\hat{\sigma}^2_{\text{mean}} = \frac{100.001}{4.001} = \mathbf{24.99}, \qquad \hat{\sigma}^2_{\text{mode}} = \frac{100.001}{6.001} = \mathbf{16.66} \qquad (\text{MLE } 200/10 = 20).$$



The posterior is right-skewed, so the mean lies above the mode.

› (c) Consistency.

Write $\hat{\sigma}^2 = \frac{b+S/2}{a+n/2+k}$ with $k = -1$ (mean) or $+1$ (mode), $S = \sum Y_i^2$. Then

$$\mathrm{E}\hat{\sigma}^2 = \frac{b + n\sigma^2/2}{a + n/2 + k} \to \sigma^2, \qquad \mathrm{Var}\hat{\sigma}^2 = \frac{2n\sigma^4/4}{(a + n/2 + k)^2} = O\Big(\frac{1}{n}\Big) \to 0,$$

so MSE $=$ bias^2 $+$ Var $\to 0$ for any fixed a, b. With $\sigma^2 = 20$: MSE(mean) $= 149.9,\ 8.5,\ 0.80$ and MSE(mode) $= 66.7,\ 7.8,\ 0.80$ for $n = 10, 100, 1000$.

Answer

(a) Mean $\frac{b+\sum Y_i^2/2}{a+n/2-1}$, mode $\frac{b+\sum Y_i^2/2}{a+n/2+1}$. (b) 24.99 and 16.66. (c) Both have bias $O(1/n)$ and variance $O(1/n)$, so their MSE $\to 0$: both are consistent, whatever a, b.

Takeaway & common mistakes

For small n the choice of point estimate matters (mean 25 vs. mode 17); asymptotically it does not.

Exercise 9.5 — Frequentist coverage of Jeffreys credible intervals

Exercise 5 (exact text from the book)

Assume $Y|\theta \sim \text{Binomial}(n, \theta)$ and $\theta \sim \text{Beta}(1/2, 1/2)$. Use a simulation study to compute the empirical coverage of the equal-tailed 95% credible set for $n \in \{1, 5, 10, 25\}$ and true value $\theta_0 \in \{0.05, 0.10, \ldots, 0.50\}$. Comment on the frequentist properties of the Bayesian credible set.

In plain words

In repeated samples, how often does the Jeffreys 95% interval contain the true success probability?

Background you need

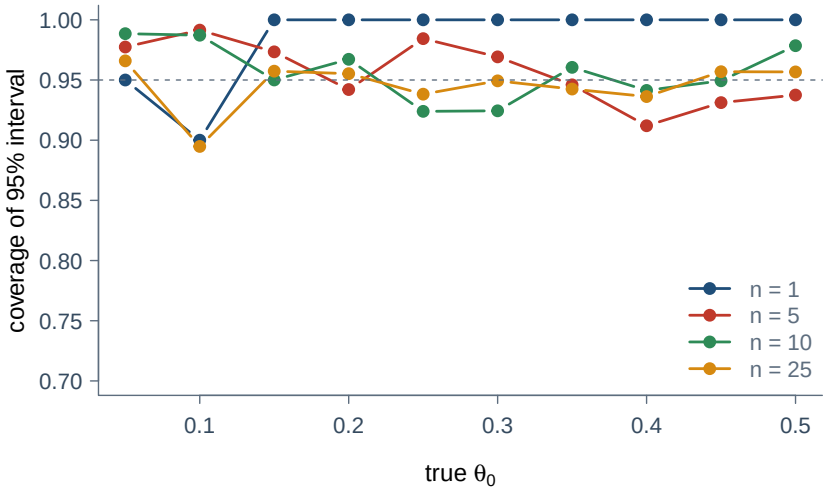
$\theta \mid Y \sim \text{Beta}(Y + \frac{1}{2},\ n - Y + \frac{1}{2})$; equal-tailed interval: its 2.5% and 97.5% quantiles. Because Y takes only $n + 1$ values, coverage can be computed *exactly*: $\sum_{y=0}^{n} \text{Binomial}(y; n, \theta_0)\, \mathbb{1}\{\theta_0 \in \text{CI}(y)\}$.

Key idea

Exact computation is the limit of an infinitely long simulation; we also ran a 10,000-replicate simulation to check.

Solution

θ_0	$n = 1$	$n = 5$	$n = 10$	$n = 25$
0.05	0.950	0.977	0.988	0.966
0.10	0.900	0.991	0.987	0.895
0.15	1.000	0.973	0.950	0.957
0.20	1.000	0.942	0.967	0.955
0.25	1.000	0.984	0.924	0.938
0.30	1.000	0.969	0.924	0.949
0.35	1.000	0.946	0.961	0.942
0.40	1.000	0.912	0.941	0.936
0.45	1.000	0.931	0.949	0.957
0.50	1.000	0.937	0.979	0.957
average	0.985	0.956	0.957	0.945



The simulation agrees with the exact values to within 0.006.

› Comments.

- On *average* over θ_0 the coverage is close to 95% even for tiny n — the Jeffreys interval has good frequentist properties (it is known to beat the classical Wald interval).
- For a *given* θ_0 the coverage jumps around (from 0.89 to 0.99 for $n = 25$) because Y is discrete: as θ_0 moves, whole outcomes enter or leave the interval. No interval for a binomial proportion can have exactly 95% coverage everywhere.
- With $n = 1$ the interval is very wide, so coverage is usually 100% (but 90% at $\theta_0 = 0.1$).

Answer

The Jeffreys equal-tailed interval has approximately nominal coverage on average (0.95–0.96 for $n \geq 5$), with oscillations between about 0.89 and 0.99 due to the discreteness of Y; for $n = 1$ it is conservative.

Takeaway & common mistakes

A Bayesian interval with a sensible default prior is often an excellent *frequentist* interval too.

Exercise 9.6 — A Bayesian re-analysis from a published estimate

Exercise 6 (exact text from the book)

A paper reports the results of a logistic regression analysis using maximum likelihood estimation. They estimate β_1 to be $\hat{\beta}_1 = 1.8$, with the 95% confidence interval (0.42, 3.17). Further, the p-value for the test that $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$ is 0.01. Conduct a Bayesian analysis using only this information and asymptotic arguments.

In plain words

We only have a point estimate, an interval and a p-value. Turn them into an approximate likelihood and do a Bayesian analysis.

Background you need

Large-sample likelihood: $\hat{\beta}_1 \mid \beta_1 \approx \mathrm{N}(\beta_1,\ \mathrm{SE}^2)$ (primer, point 3); a 95% CI is $\hat{\beta} \pm 1.96\,\mathrm{SE}$. Normal prior $\mathrm{N}(0, \tau^2) \Rightarrow$ posterior $\mathrm{N}\big(\frac{\hat{\beta}/\mathrm{SE}^2}{1/\mathrm{SE}^2+1/\tau^2},\ \frac{1}{1/\mathrm{SE}^2+1/\tau^2}\big)$. Point-null BF: $\frac{\mathrm{N}(\hat{\beta};0,\mathrm{SE}^2+\tau^2)}{\mathrm{N}(\hat{\beta};0,\mathrm{SE}^2)}$ (as in equation 5.10).

Key idea

Recover the SE from the interval, check it against the p-value, then combine the normal "likelihood" with a few priors.

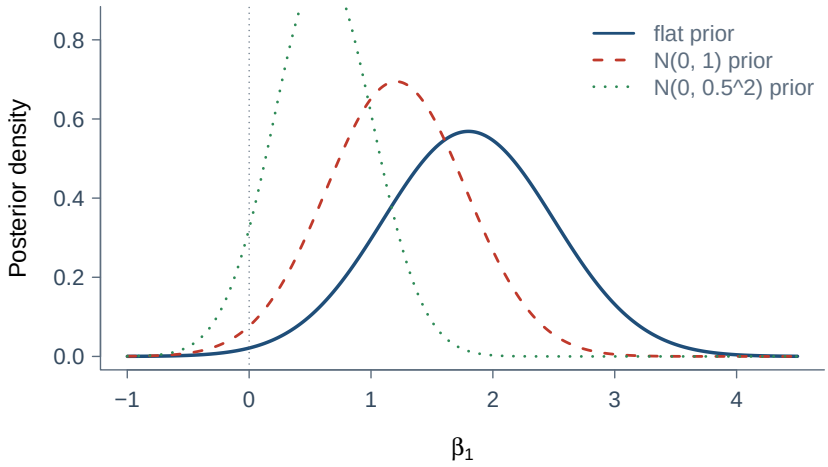
Solution

Step 1 — the standard error.

$\mathrm{SE} = \frac{3.17-0.42}{2\times 1.96} = 0.70$. Check: $z = 1.8/0.70 = 2.57$, two-sided $p = 0.010$ — consistent with the reported $p = 0.01$ (and the interval is symmetric about 1.8, as a Wald interval should be).

Step 2 — posteriors under several priors.

Prior on β_1	posterior mean	95% CI	$\mathrm{Prob}(\beta_1 > 0)$	mean odds ratio	$\mathrm{BF}(H_1{:}H_0)$
flat	1.80	(0.43, 3.17)	0.995	7.7	—
$\mathrm{N}(0, 2^2)$	1.60	(0.31, 2.90)	0.992	6.2	6.2
$\mathrm{N}(0, 1^2)$	1.21	(0.08, 2.33)	0.982	3.9	5.2
$\mathrm{N}(0, 0.5^2)$ (sceptical)	0.61	$(-0.19,\ 1.40)$	0.932	2.0	2.5



Why? With a flat prior the posterior reproduces the confidence interval numerically, but now it may be read as "95% probability that β_1 lies in (0.43, 3.17)". The mean odds ratio is $\mathrm{E}(e^{\beta_1}) = e^{m+s^2/2}$ for a normal posterior.

Answer

$\beta_1 \mid \text{data} \approx \mathrm{N}(1.8,\ 0.70^2)$ under a flat prior: $\mathrm{Prob}(\beta_1 > 0) = 0.995$, 95% interval (0.43, 3.17). Reasonable sceptical priors shrink the estimate (e.g. to 1.21 with $\mathrm{N}(0, 1)$) but keep $\mathrm{Prob}(\beta_1 > 0) \geq 0.93$. The point-null Bayes factor is only 2.5–6.2 — much weaker evidence than "$p = 0.01$" suggests.

Takeaway & common mistakes

A p-value of 0.01 corresponds to a Bayes factor of only about 3–6 against the point null: p-values overstate the evidence against H_0.

Exercise 9.7 — How many customers for a posterior SD of 0.01?

Exercise 7 (exact text from the book)

You are designing a study to estimate the success probability of a new marketing strategy. When the data have been collected, you will analyze them using the model $Y|\theta \sim \text{Binomial}(n, \theta)$ and $\theta \sim \text{Uniform}(0, 1)$. Before collecting any data, you suspect (based on past studies) that $\theta \approx 0.4$. Which value of n should be used to ensure the posterior standard deviation will be approximately 0.01?

In plain words

Choose the sample size so that, after the study, we will know the success rate to within about ± 0.01 (one posterior SD).

Background you need

$\theta \mid Y \sim \text{Beta}(Y+1,\ n-Y+1)$, whose variance is $\frac{ab}{(a+b)^2(a+b+1)}$ with $a+b=n+2$. If $Y \approx n\theta$ this is $\approx \frac{\theta(1-\theta)}{n+3}$.

Key idea

Plug in the guess $\theta = 0.4$ and solve for n; then check by averaging the exact posterior SD over the possible outcomes Y.

Solution

Approximation.

$$\sqrt{\frac{0.4\times 0.6}{n+3}} = 0.01 \iff n+3 = \frac{0.24}{0.0001} = 2400 \iff n = 2397.$$

Exact check.

Averaging the exact Beta posterior SD over $Y \sim \text{Binomial}(2397, 0.4)$ gives 0.01000, and 2397 is the smallest n for which it is ≤ 0.01.

Sensitivity to the guess.

If θ were 0.3 the SD at $n = 2397$ would be 0.0094; if 0.5, 0.0102 — the worst case $\theta = 0.5$ would need $n = 2497$.

Answer

$n \approx 2{,}400$ (precisely $n = 2397$ under $\theta \approx 0.4$; use $n \approx 2{,}500$ to be safe for any θ).

Takeaway & common mistakes

Posterior precision grows like $\sqrt{n}$: halving the posterior SD needs four times as many observations.

Exercise 9.8 — Exponential families, conjugacy and unbiasedness

Exercise 8 (exact text from the book)

Suppose $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} f(y|\eta)$ where f is the canonical exponential family

$$f(y|\eta) \propto \exp\{y\eta - \psi(\eta)\}$$

for some known function ψ. We are interested in estimating the parameter $\theta = E(Y_i|\eta) = d\psi(\eta)/d\eta$.

(a) Show that the Gaussian density with variance fixed at one and the Poisson density can be written in this form.

(b) Obtain the maximum likelihood estimator of θ. Is it unbiased for θ?

(c) Find a class of conjugate priors for η.

(d) Obtain the Bayes estimator of θ using a conjugate prior and squared error loss.

(e) Is there any (proper) conjugate prior for which the Bayes estimator that you obtained in (d) is unbiased? Justify your answer.

In plain words

For the whole exponential family at once: write two familiar models in canonical form, find the MLE and the conjugate prior, and show that the Bayes estimator can never be unbiased under a proper conjugate prior.

Background you need

Primer, point 4: $\mathrm{E}(Y \mid \eta) = \psi'(\eta)$. **Diaconis–Ylvisaker (1979)**: for the conjugate prior below, the prior mean of $\psi'(\eta)$ is y_0, and the posterior mean of $\psi'(\eta)$ is a weighted average of y_0 and $\bar{Y}$.

Key idea

The conjugate posterior mean is $\frac{n_0 y_0 + n\bar{Y}}{n_0+n}$: it always puts weight $\frac{n_0}{n_0+n} > 0$ on the prior guess y_0, which creates bias.

Solution

(a) Gaussian, variance 1.

$f(y \mid \mu) \propto \exp\{-\frac{(y-\mu)^2}{2}\} = \exp\{y\mu - \frac{\mu^2}{2}\}\, e^{-y^2/2}$: $\eta = \mu$, $\psi(\eta) = \frac{\eta^2}{2}$ (the factor $e^{-y^2/2}$ does not involve η). Check: $\psi'(\eta) = \eta = \mu = \mathrm{E}Y$. ✓

Poisson. $f(y \mid \lambda) = \frac{e^{-\lambda}\lambda^y}{y!} = \exp\{y \log\lambda - \lambda\}\frac{1}{y!}$: $\eta = \log\lambda$, $\psi(\eta) = e^\eta$. Check: $\psi'(\eta) = e^\eta = \lambda = \mathrm{E}Y$. ✓

(b) MLE.

$\ell(\eta) = \eta\sum Y_i - n\psi(\eta)$, so $\ell'(\eta) = \sum Y_i - n\psi'(\eta) = 0 \Rightarrow \psi'(\hat{\eta}) = \bar{Y}$. By invariance, $\hat{\theta}_{\text{MLE}} = \psi'(\hat{\eta}) = \bar{Y}$, which is unbiased: $\mathrm{E}\bar{Y} = \theta$.

(c) Conjugate prior.

Take a prior with the same shape as the likelihood in η:

$$\pi(\eta \mid n_0, y_0) \propto \exp\{n_0 y_0 \eta - n_0\psi(\eta)\}, \qquad n_0 > 0,$$

(like n_0 prior observations with mean y_0). Then

$$p(\eta \mid \mathbf{Y}) \propto \exp\{(n_0 y_0 + n\bar{Y})\eta - (n_0 + n)\psi(\eta)\} :$$

the same family with $n_0 \to n_0 + n$ and $y_0 \to \frac{n_0 y_0 + n\bar{Y}}{n_0 + n}$.

› (d) Bayes estimator under squared-error loss

= posterior mean of $\theta = \psi'(\eta)$. For this family (Diaconis–Ylvisaker) $\mathrm{E}\{\psi'(\eta)\} = y_0$ under the prior $\pi(\cdot \mid n_0, y_0)$; applying this to the posterior,

$$\hat{\theta}_{\text{Bayes}} = \mathrm{E}\{\psi'(\eta) \mid \mathbf{Y}\} = \frac{n_0 y_0 + n\bar{Y}}{n_0 + n}.$$

Why? The prior result follows by integrating $\frac{\mathrm{d}}{\mathrm{d}\eta}\pi(\eta) = n_0\{y_0 - \psi'(\eta)\}\pi(\eta)$ over η: the left side integrates to 0 (the density vanishes at the ends of its range), so $\mathrm{E}\{\psi'(\eta)\} = y_0$.

For example, normal: $\frac{n_0 y_0 + n\bar{Y}}{n_0 + n}$ is the normal–normal posterior mean; Poisson: $\lambda \sim \mathrm{Gamma}(n_0 y_0, n_0)$ gives $\frac{n_0 y_0 + \sum Y_i}{n_0 + n}$.

› (e) Can it be unbiased?

$$\mathrm{E}\big(\hat{\theta}_{\text{Bayes}} \mid \theta\big) = \frac{n_0 y_0 + n\theta}{n_0 + n} = \theta \iff n_0(y_0 - \theta) = 0.$$

For this to hold for *every* θ we need $n_0 = 0$, but then $\pi(\eta) \propto 1$, which is improper. So **no proper conjugate prior gives an unbiased Bayes estimator** (except trivially at the single point $\theta = y_0$).

Answer

(a) Normal: $\eta = \mu$, $\psi = \eta^2/2$; Poisson: $\eta = \log\lambda$, $\psi = e^\eta$. (b) $\hat{\theta} = \bar{Y}$, unbiased. (c) $\pi(\eta) \propto \exp\{n_0 y_0 \eta - n_0 \psi(\eta)\}$. (d) $\hat{\theta}_{\text{Bayes}} = \frac{n_0 y_0 + n\bar{Y}}{n_0 + n}$. (e) No: unbiasedness for all θ requires $n_0 = 0$, an improper prior.

Takeaway & common mistakes

This is a special case of a general theorem: under squared-error loss, a Bayes estimator with a proper prior is never unbiased (unless it estimates θ perfectly). Bayesian estimators deliberately accept bias for lower MSE (Exercise 3).